



AI-ENHANCED WASTE COLLECTION

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OUR CLIENT

Recycle**n**XT

A company on a mission to
redefine how the world thinks
about waste by turning trash
into valuable resources.



THE PROJECT

We are optimising the waste collection process to reduce the environmental impacts and make waste collection more accessible.

USER STORIES



01

Waste Collectors

As a **waste collector**, I want to be able to **view the routes** easily so that I can **follow them and collect the waste**

02

Waste Management Company/City Council

As a **city council/municipal**, I want to be able to **enter the collection points and get the most efficient route to save money and time.**



04

Households/Businesses

As a **household/business**, I want an **appropriate frequency of waste collection** so that the **bins are not overflowing.**



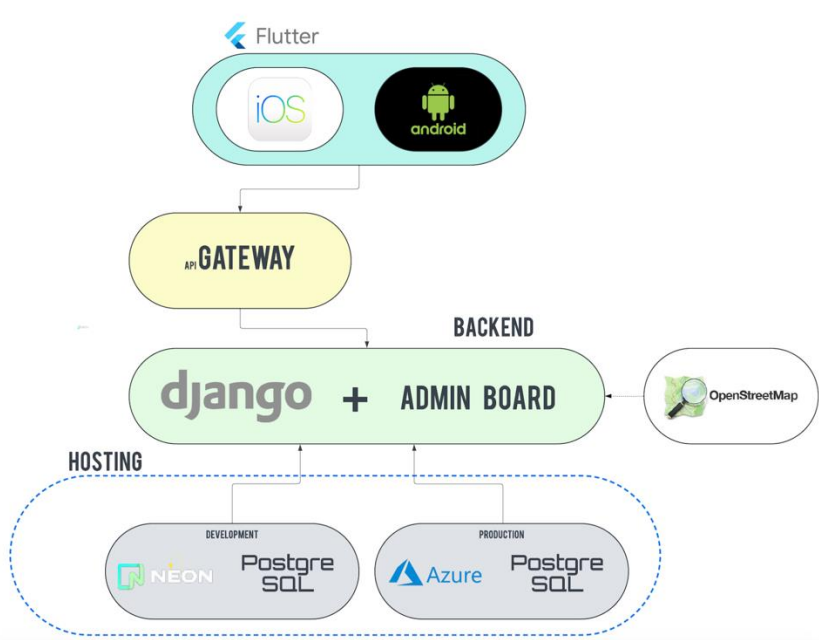
03

Environmental Organization

As an **environmental organization**, I want to be able to **see that the routes are reducing the amount of carbon and emissions via the report**, so I can be **sure the company is helping to protect the environment.**



ARCHITECTURE



Why we chose our architecture:

General:

- Previous Project Inspiration
- Mentor Guidance

Specific:

Flutter

Consistent look across platforms: Flutter's single codebase can build apps for iOS and Android.

Highly customized UIs: Flutter's flexibility and control with a tree-like structure make it easy for building and managing interactive UIs.

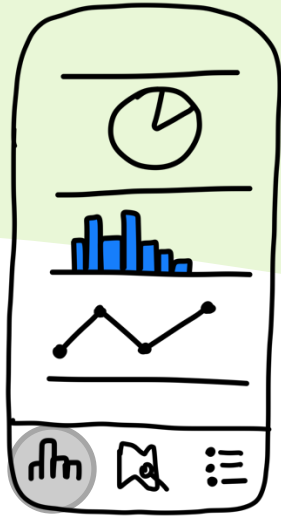
Django

Scalability: Django's built-in features such as caching and load balancing allow it to scale well which was a main criteria presented by our client.

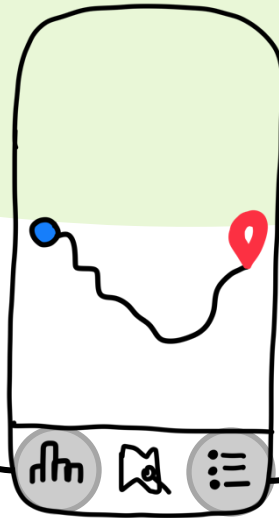
API's: Django ships with support for REST API's which allow for easy web development and secure data exchange



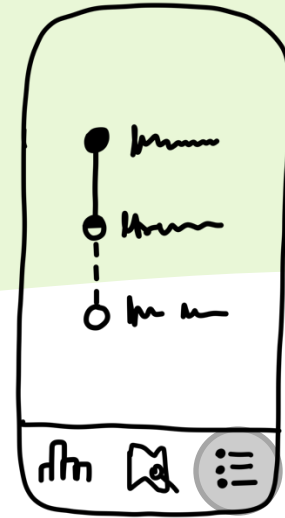
INTERFACE DESIGN



Metrics Page



Map Page



Route Page



TESTING STRATEGY



WIDGET TESTS

Widget tests allow us to testing the app's UI components to check they are responding as expected,



UNIT TESTS

We are using unit testing to make sure the functionality of the code is working as expected.



CONTINUOUS INTEGRATION

Our continuous integration runs our widget and unit tests then builds the flutter application on ubuntu and iOS.



CONTINUOUS DEPLOYMENT

We are planning on deploying using Azure as this is what our client specified in his criteria.



OUR PLANS FOR USER TESTING

01

Usability Testing

Record users interacts with the app to test and optimise the usability of the interface.

02

Questionnaires

Develop or use a pre defined questionnaire to gather tester feedback.



DEMO



10:26

ReCycleNXT

Welcome Back!

Username

Password

[Forgot Password?](#)

Sign In

Not a member? [Register Here.](#)





OUR PROCESS

WEEKLY MEETINGS

Monday and Wednesday meetings to check progress and decide what to do next.

DISCORD

Using a discord chat with channels to organise our conversations.

TASKS + SPRINTS

Separating jobs into manageable tasks and organising them into sprints

GITHUB

Implementing templates early on to ensure detailed issues and pull requests.

CHALLENGES



Evolving Requirements

- **Challenge:** Frequent client expectation changes delayed implementation.
- **Solution:** Agile sprints ensured quick alignment with updates.



Resource Constraints

- **Challenge:** Limited client feedback slowed decisions.
- **Solution:** Used indirect communication and temporary predictions to keep the project moving forward.



SET UP

- **Challenge:** Flutter & Android SDK setup caused delays.
- **Solution:** Created setup guides for smoother onboarding.

USER FEEDBACK

- Whilst we have met our client for the MVP hand over we are still waiting on feedback as the demo is being handed over to the team.





THANK YOU!

